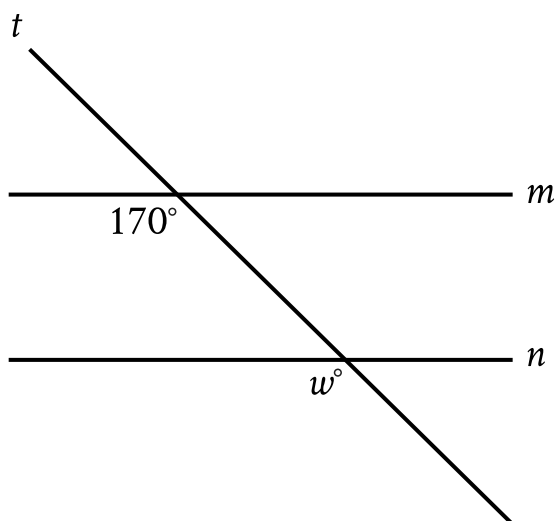


ID: 9912e19f

Triangles EFG and JKL are congruent, where E , F , and G correspond to J , K , and L , respectively. The measure of angle E is 45° and the measure of angle F is 20° . What is the measure of angle J ?

- A. 20°
- B. 45°
- C. 135°
- D. 160°

ID: 5207e508



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

- A. 17
- B. 30
- C. 70
- D. 170

ID: 93f48423

What is the area, in square inches, of a rectangle with a length of **7** inches and a width of **6** inches?

- A. **13**
- B. **20**
- C. **42**
- D. **84**

ID: 5252e606

The side length of a square is **55 centimeters (cm)**. What is the area, **in cm^2** , of the square?

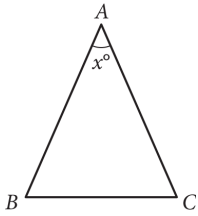
- A. 110
- B. 220
- C. 3,025
- D. 12,100

ID: bb560789

Triangle R has an area of **80 square centimeters (cm^2)**. Square S has side lengths of **4 cm**. What is the total area of triangle R and square S, **in cm^2** ?

- A. **42**
- B. **44**
- C. **84**
- D. **96**

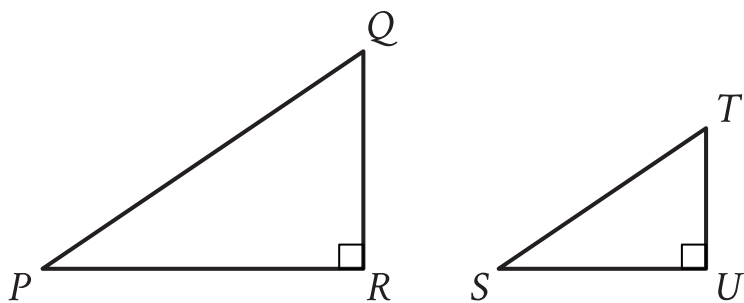
ID: c8d60e48



In the given triangle, $AB = AC$ and $\angle ABC$ has a measure of 67° . What is the value of x ?

- A. 36
- B. 46
- C. 58
- D. 70

ID: d5f349b7

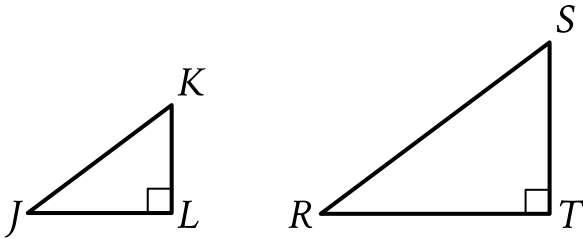


Note: Figures not drawn to scale.

Right triangles PQR and STU are similar, where P corresponds to S . If the measure of angle Q is 18° , what is the measure of angle S ?

- A. 18°
- B. 72°
- C. 82°
- D. 162°

ID: babd7461

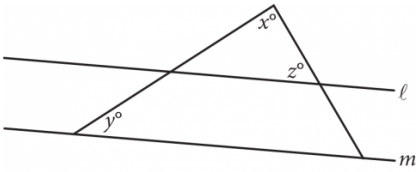


Note: Figure not drawn to scale.

In the figure shown, triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . The length of $\overline{JK}$ is 15, and the perimeter of triangle JKL is 36. The length of $\overline{RS}$ is 135. What is the perimeter of triangle RST ?

- A. 324
- B. 540
- C. 2,916
- D. 8,100

ID: a6dbad6b



Note: Figure not drawn to scale.

In the figure above, lines ℓ and m are parallel, $y = 20$, and $z = 60$. What is the value of x ?

- A. 120
- B. 100
- C. 90
- D. 80

ID: 59cb654c

The area of a square is **64** square inches. What is the side length, in inches, of this square?

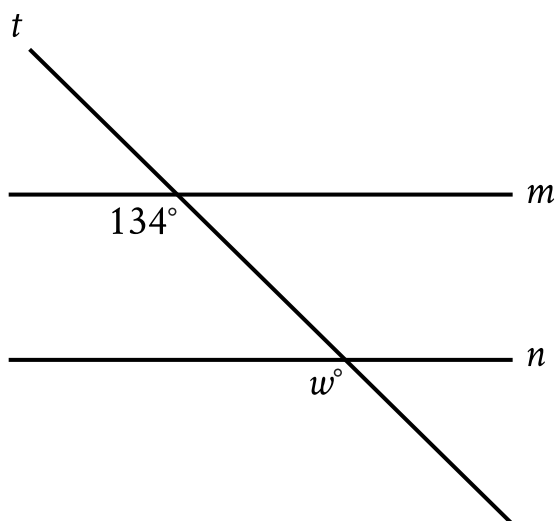
- A. 8
- B. 16
- C. **64**
- D. 128

ID: cbe8ca31

In $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . What is the measure of $\angle Z$?

- A. 58°
- B. 74°
- C. 122°
- D. 212°

ID: c24e1bda

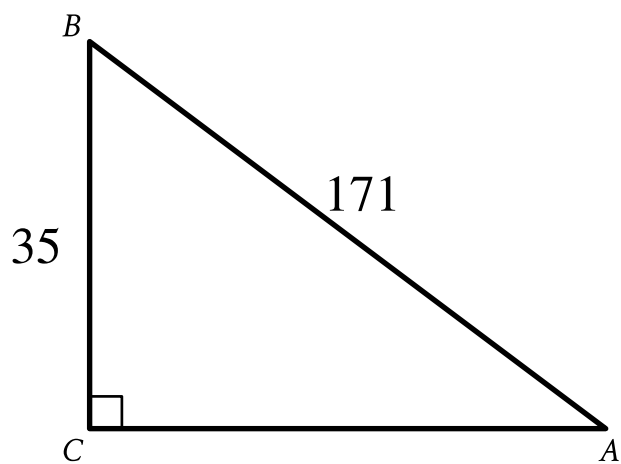


Note: Figure not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

- A. 13
- B. 34
- C. 66
- D. 134

ID: 87a9a2d4

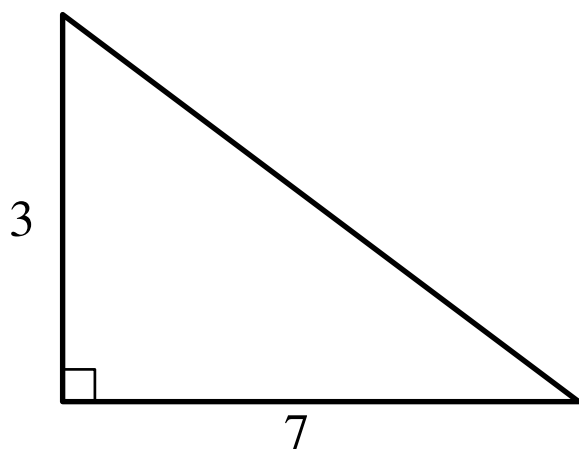


Note: Figure not drawn to scale.

In the right triangle shown, what is the value of $\sin A$?

- A. $\frac{1}{171}$
- B. $\frac{35}{171}$
- C. $\frac{171}{35}$
- D. 171

ID: e6f2ace7

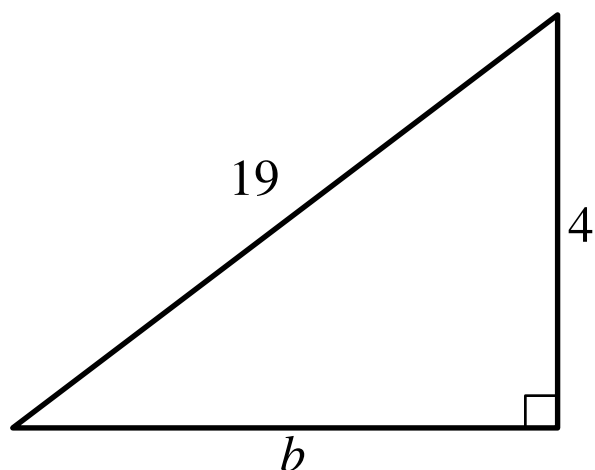


Note: Figure not drawn to scale.

The lengths of the legs of a right triangle are shown. Which of the following is closest to the length of the triangle's hypotenuse?

- A. 3.2
- B. 5
- C. 7.6
- D. 20

ID: b0c5ece5



Note: Figure not drawn to scale.

Which equation shows the relationship between the side lengths of the given triangle?

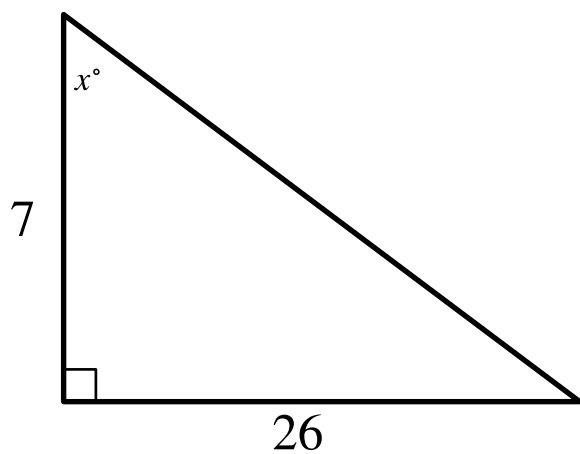
- A. $4b = 19$
- B. $4 + b = 19$
- C. $4^2 + b^2 = 19^2$
- D. $4^2 - b^2 = 19^2$

ID: d0b6d927

A rectangle has an area of **63** square meters and a length of **9** meters. What is the width, in meters, of the rectangle?

- A. **7**
- B. **54**
- C. **81**
- D. **567**

ID: 64c1f044

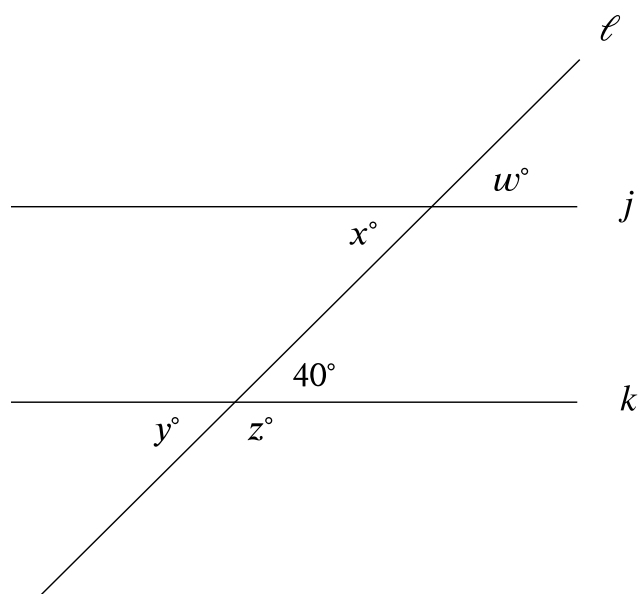


Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\tan x^\circ$?

- A. $\frac{1}{26}$
- B. $\frac{19}{26}$
- C. $\frac{26}{7}$
- D. $\frac{33}{7}$

ID: 9d078710



Note: Figure not drawn to scale.

In the figure shown, line ℓ intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

- A. $w = 40$
- B. $x = 140$
- C. $y = 40$
- D. $z = 140$

ID: 0e40dfb0

A rectangle has a length of **3** units and a width of **39** units. Which expression gives the area, in square units, of this rectangle?

A. $2(3 + 39)$

B. $2(3 \cdot 39)$

C. $3 + 39$

D. $3 \cdot 39$

ID: 0d43db90

The perimeter of triangle ABC is **17** inches, the length of side AB is **4** inches, and the length of side AC is **7** inches. What is the length, in inches, of side BC ?

- A. **4**
- B. **6**
- C. **7**
- D. **11**

ID: fc5ef8d3

The table gives the perimeters of similar triangles TUV and XYZ , where $\overline{TU}$ corresponds to $\overline{XY}$. The length of $\overline{TU}$ is 6.

	Perimeter
Triangle TUV	50
Triangle XYZ	150

What is the length of $\overline{XY}$?

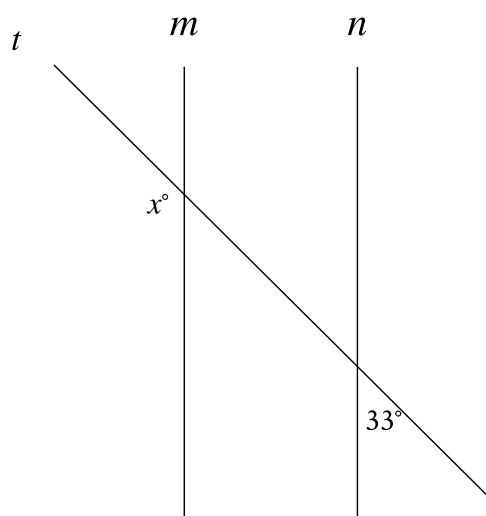
- A. 2
- B. 6
- C. 18
- D. 56

ID: f9d40000

In $\triangle XYZ$, the measure of $\angle X$ is 23° and the measure of $\angle Y$ is 66° . What is the measure of $\angle Z$?

- A. 43°
- B. 89°
- C. 91°
- D. 179°

ID: 0d3f51dc



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n , and line t intersects both lines. What is the value of x ?

- A. 33
- B. 57
- C. 123
- D. 147

ID: b434e103

In $\triangle RST$, the measure of $\angle R$ is 63° . Which of the following could be the measure, in degrees, of $\angle S$?

- A. 116
- B. 118
- C. 126
- D. 180

ID: 689abc2a

Rectangle P has an area of **72** square inches. If a rectangle with an area of **20** square inches is removed from rectangle P, what is the area, in square inches, of the resulting figure?

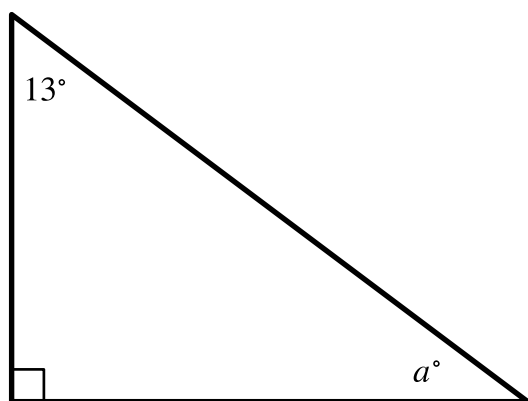
- A. **92**
- B. **84**
- C. **80**
- D. **52**

ID: bbaac300

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D , and angles C and F are right angles. If $\cos B = \frac{1}{22}$, what is the value of $\cos E$?

- A. $\frac{1}{22}$
- B. $\frac{1}{23}$
- C. $\frac{21}{22}$
- D. $\frac{22}{23}$

ID: 69f4bbdc



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of a ?

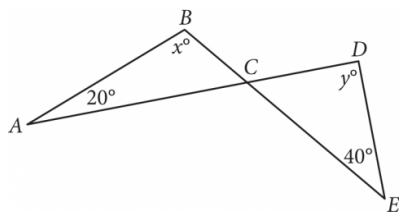
- A. 13
- B. 77
- C. 90
- D. 103

ID: 3563d76d

At a certain time and day, the Washington Monument in Washington, DC, casts a shadow that is 300 feet long. At the same time, a nearby cherry tree casts a shadow that is 16 feet long. Given that the Washington Monument is approximately 555 feet tall, which of the following is closest to the height, in feet, of the cherry tree?

- A. 10
- B. 20
- C. 30
- D. 35

ID: dfc420b2



Note: Figure not drawn to scale.

In the figure above, $\overline{AD}$ intersects $\overline{BE}$ at C . If

$x = 100$, what is the value of y ?

- A. 100
- B. 90
- C. 80
- D. 60

ID: f2495de4

What is the value of $\cos \frac{565\pi}{6}$?

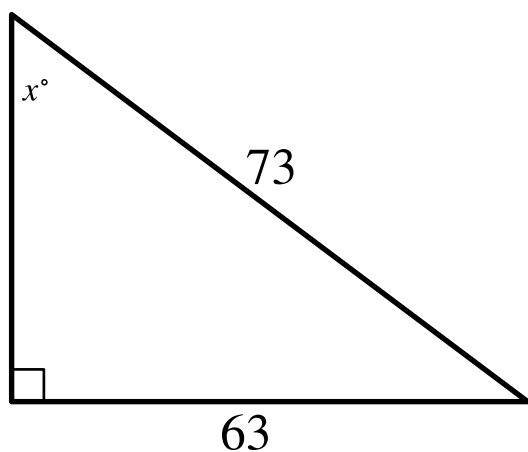
- A. $\frac{1}{2}$
- B. 1
- C. $\frac{\sqrt{3}}{2}$
- D. $\sqrt{3}$

ID: 0837c3b9

Triangle ABC and triangle DEF are similar triangles, where $\overline{AB}$ and $\overline{DE}$ are corresponding sides. If $\overline{DE} = 2\overline{AB}$ and the perimeter of triangle ABC is 20, what is the perimeter of triangle DEF ?

- A. 10
- B. 40
- C. 80
- D. 120

ID: a6097ec2



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of $\sin x^\circ$?

- A. $\frac{1}{73}$
- B. $\frac{10}{73}$
- C. $\frac{63}{73}$
- D. $\frac{136}{73}$

ID: 575f1e12

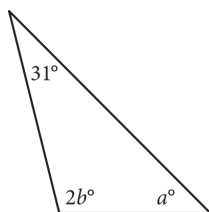
What is the area, in square centimeters, of a rectangle with a length of **34 centimeters (cm)** and a width of **29 cm**?

ID: b4767bef

The lengths of two sides of a triangle are **4** centimeters and **6** centimeters. If the perimeter of the triangle is **18** centimeters, what is the length, in centimeters, of the third side of this triangle?

- A. **2**
- B. **8**
- C. **10**
- D. **24**

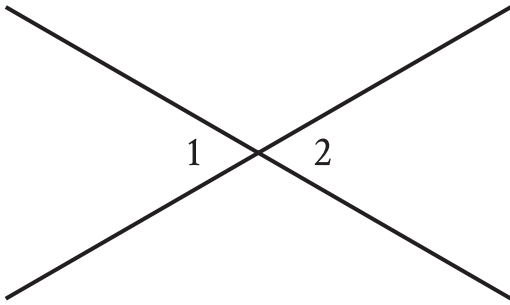
ID: 410bdb6



In the triangle above, $a = 45$. What is the value of b ?

- A. 52
- B. 59
- C. 76
- D. 104

ID: a456f28c

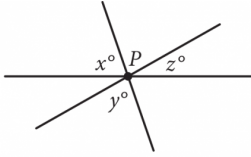


Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. Angle **1** and angle **2** are vertical angles. The measure of angle **1** is 72° . What is the measure of angle **2**?

- A. 72°
- B. 108°
- C. 144°
- D. 288°

ID: 087cdcfd



Note: Figure not drawn to scale.

In the figure, three lines intersect at point P . If $x = 65$ and $y = 75$, what is the value of z ?

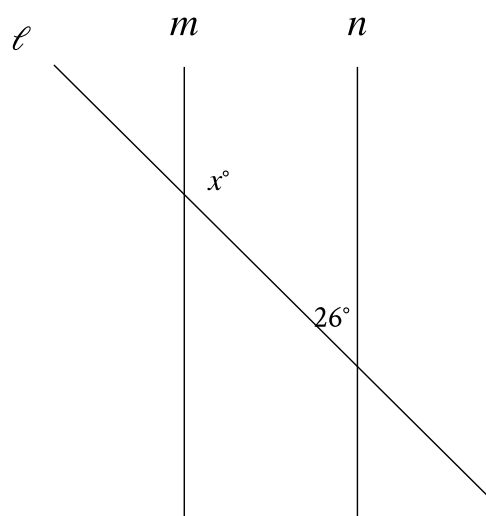
- A. 140
- B. 80
- C. 40
- D. 20

ID: c88183f7

A rectangle has a length of **13** and a width of **6**. What is the perimeter of the rectangle?

- A. **12**
- B. **26**
- C. **38**
- D. **52**

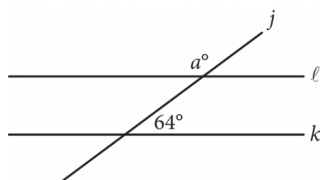
ID: afa3c48b



Note: Figure not drawn to scale.

In the figure shown, line m is parallel to line n . What is the value of x ?

- A. 13
- B. 26
- C. 52
- D. 154

ID: 992f4e93

Note: Figure not drawn to scale.

In the figure above, lines ℓ and k are parallel.

What is the value of a ?

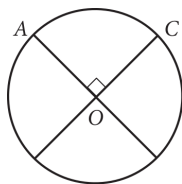
- A. 26
- B. 64
- C. 116
- D. 154

ID: f1747a6a

In triangle ABC , the measure of angle B is 52° and the measure of angle C is 17° . What is the measure of angle A ?

- A. 21°
- B. 35°
- C. 69°
- D. 111°

ID: 23c5fcce

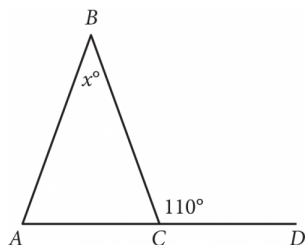


The circle above with center O has a circumference of 36.

What is the length of minor arc $\overline{AC}$?

- A. 9
- B. 12
- C. 18
- D. 36

ID: 5733ce30



In the given figure, $\overline{AC}$ extends to point D . If the measure of $\angle BAC$ is equal to the measure of $\angle BCA$, what is the value of x ?

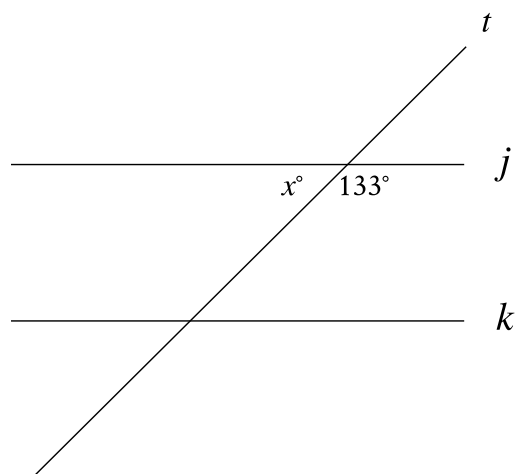
- A. 110
- B. 70
- C. 55
- D. 40

ID: 50774285

Each base of a right rectangular prism has a length of **19** inches and a width of **8** inches. The prism has a volume of **2,736** cubic inches. What is the height, in inches, of the prism?

- A. **18**
- B. **27**
- C. **144**
- D. **152**

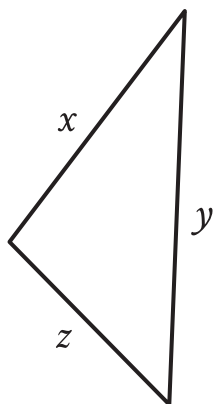
ID: 3b4b5b1e



Note: Figure not drawn to scale.

In the figure, line j is parallel to line k . What is the value of x ?

ID: 29e9b28c

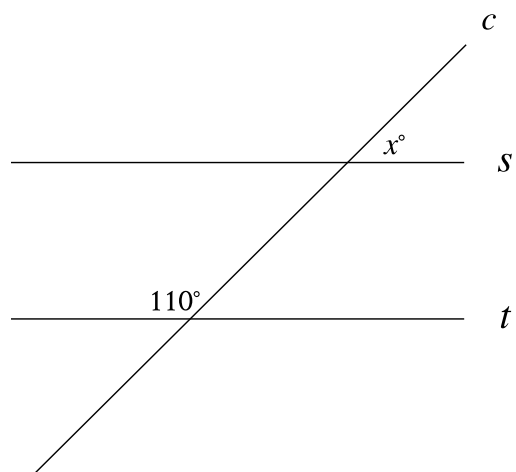


Note: Figure not drawn to scale.

The triangle shown has a perimeter of **22** units. If $x = 9$ units and $y = 7$ units, what is the value of z , in units?

- A. **6**
- B. **7**
- C. **9**
- D. **16**

ID: cf0d3050



Note: Figure not drawn to scale.

In the figure shown, line c intersects parallel lines s and t . What is the value of x ?

ID: 0bb39de4

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 18° . What is the measure of angle F ?

- A. 18°
- B. 72°
- C. 90°
- D. 162°

ID: 379ffefb

A right triangle has legs with lengths of **11** centimeters and **9** centimeters. What is the length of this triangle's hypotenuse, in centimeters?

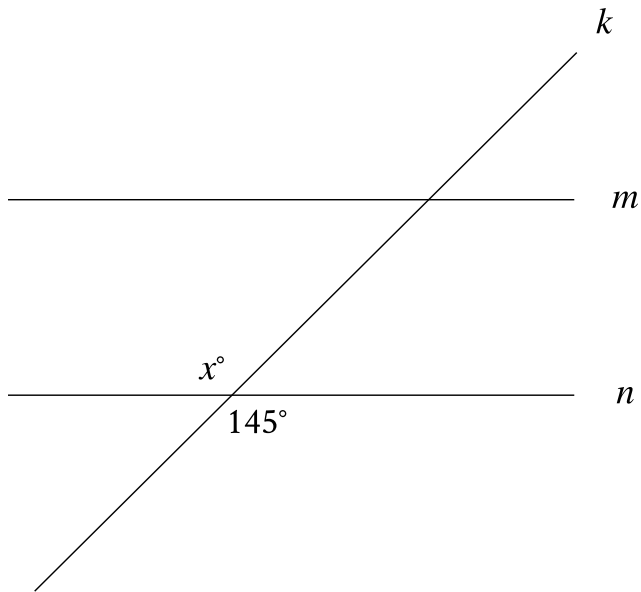
- A. $\sqrt{40}$
- B. $\sqrt{202}$
- C. **20**
- D. **202**

ID: 3453aafc

What is the area, in square centimeters, of a rectangle with a length of **36** centimeters and a width of **34** centimeters?

- A. **70**
- B. **140**
- C. **1,156**
- D. **1,224**

ID: 43236565



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n , and line k intersects both lines. Which of the following statements is true?

- A. The value of x is less than **145**.
- B. The value of x is greater than **145**.
- C. The value of x is equal to **145**.
- D. The value of x cannot be determined.

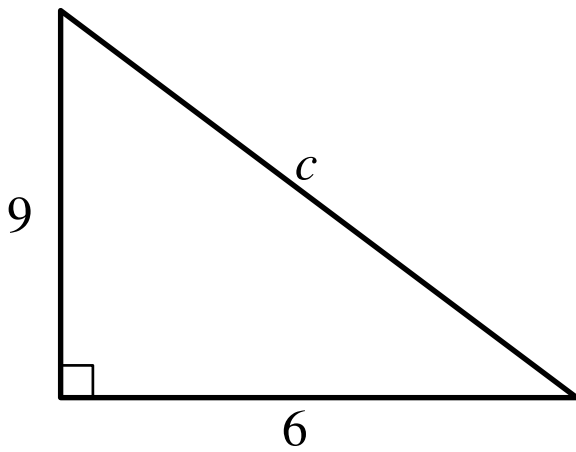
ID: d03e29f1

$$(x - 6)^2 + (y - 3)^2 = 81$$

The graph of the given equation in the xy -plane is a circle. What is the length of the radius of this circle?

- A. **3**
- B. **6**
- C. **9**
- D. **81**

ID: 36661021

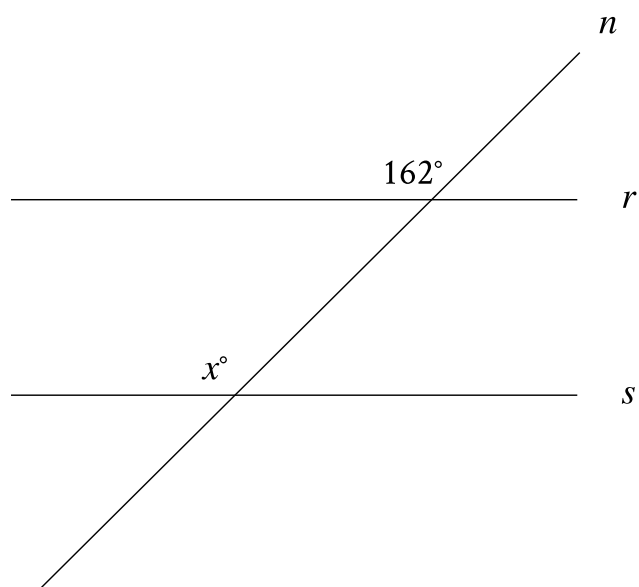


Note: Figure not drawn to scale.

In the right triangle shown, which of the following is closest to the value of c ?

- A. 7.5
- B. 10.8
- C. 15
- D. 58.5

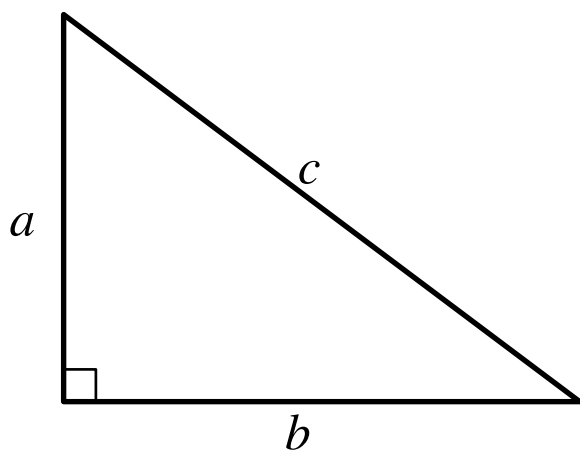
ID: 5b918ebb



Note: Figure not drawn to scale.

In the figure, line n intersects lines r and s . Line r is parallel to line s . What is the value of x ?

ID: c9f8d1e9



Note: Figure not drawn to scale.

For the right triangle shown, $a = 4$ and $b = 5$. Which expression represents the value of c ?

- A. $4 + 5$
- B. $\sqrt{(4)(5)}$
- C. $\sqrt{4 + 5}$
- D. $\sqrt{4^2 + 5^2}$

ID: f60bb551

The area of a rectangle is **630** square inches. The length of the rectangle is **70** inches. What is the width, in inches, of this rectangle?

- A. **9**
- B. **70**
- C. **315**
- D. **560**

ID: d2047497

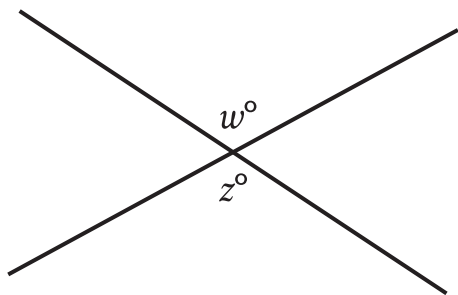
What is the area of a rectangle with a length of **17 centimeters (cm)** and a width of **7 cm**?

- A. **24 cm²**
- B. **48 cm²**
- C. **119 cm²**
- D. **576 cm²**

ID: 4efea6a3

The area of a rectangle is **57** square inches. The length of the longest side of the rectangle is **19** inches. What is the length, in inches, of the shortest side of this rectangle?

ID: 64d1f49f



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $w = 136$, what is the value of z ?

- A. 36
- B. 44
- C. 68
- D. 136

ID: f39f88b7

A triangle has a base length of **40** centimeters and a height of **90** centimeters. What is the area, in square centimeters, of the triangle?

ID: 76670c80

Each side of a square has a length of 45. What is the perimeter of this square?

ID: 4420e500

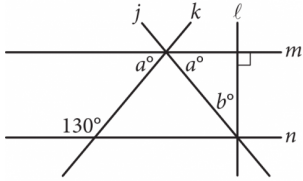
What is the area of a rectangle with a length of **4 centimeters (cm)** and a width of **2 cm**?

- A. **6 cm²**
- B. **8 cm²**
- C. **12 cm²**
- D. **36 cm²**

ID: 165c30c4

A rectangle has a length of **64** inches and a width of **32** inches. What is the area, in square inches, of the rectangle?

ID: 3828f53d



Note: Figure not drawn to scale.

In the figure above, lines m and n are parallel.

What is the value of b ?

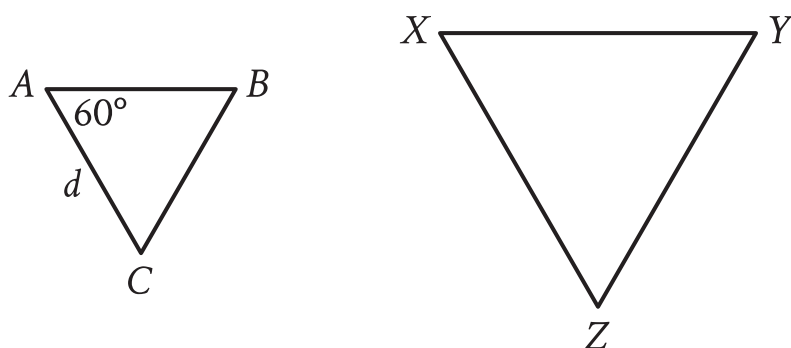
- A. 40
- B. 50
- C. 65
- D. 80

ID: 739f1bbc

In triangle ABC , $AB = 4,680$ millimeters (mm) and $BC = 4,680$ mm. Which statement is sufficient to prove that triangle ABC is equilateral?

- A. $AC = 4,680$ mm
- B. $AC = 468$ mm
- C. $AC = 46.8$ mm
- D. $AC = 4.68$ mm

ID: e0d2e21a



Note: Figures not drawn to scale.

For the triangles shown, triangle ABC is dilated by a scale factor of **3** to obtain triangle XYZ , where $d = 16$. What is the measure, in degrees, of angle X ?

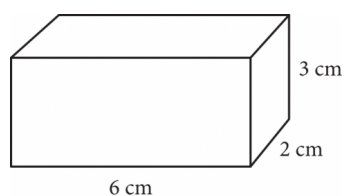
- A. **20**
- B. **57**
- C. **60**
- D. **63**

ID: 42b4493b

In a right triangle, the measure of one of the acute angles is 51° . What is the measure, in degrees, of the other acute angle?

- A. 6
- B. 39
- C. 49
- D. 51

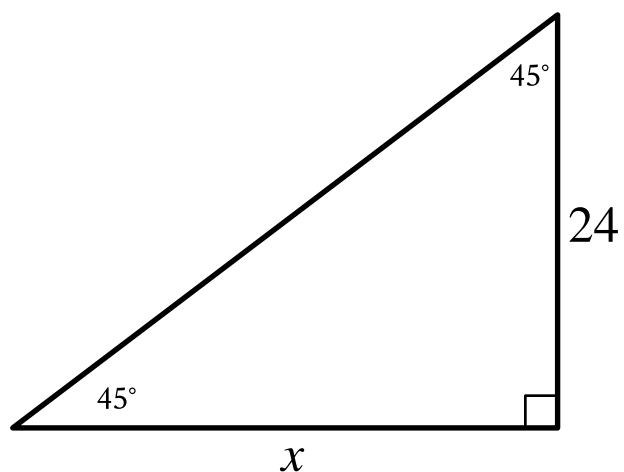
ID: d683a9cc



The figure shows the lengths, in centimeters (cm), of the edges of a right rectangular prism. The volume V of a right rectangular prism is ℓwh , where ℓ is the length of the prism, w is the width of the prism, and h is the height of the prism. What is the volume, in cubic centimeters, of the prism?

- A. 36
- B. 24
- C. 12
- D. 11

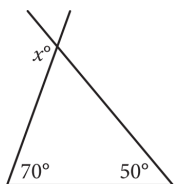
ID: 145337bc



Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

- A. 24
- B. 45
- C. 48
- D. 69

ID: 36200a38

In the figure above, two sides of a triangle are extended. What is the value of x ?

- A. 110
- B. 120
- C. 130
- D. 140

ID: a490003a

The width of a rectangle is **7** centimeters. The length of the rectangle is **40** centimeters longer than the width. What is the area, in square centimeters, of this rectangle?

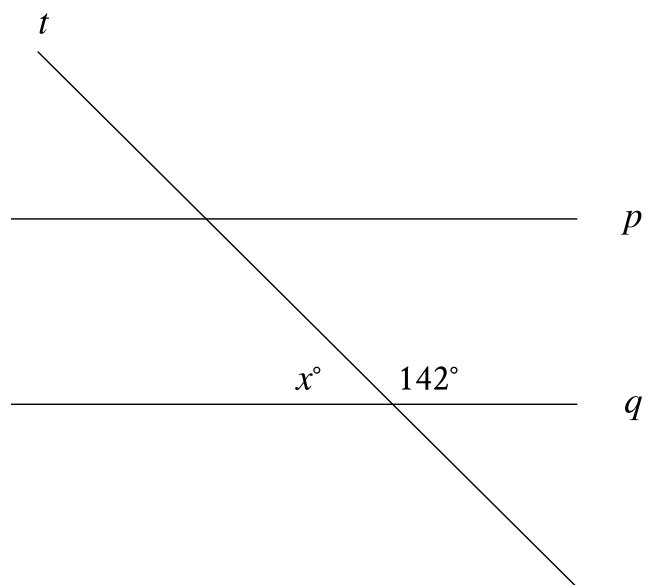
- A. **7**
- B. **14**
- C. **54**
- D. **329**

ID: 02b02213

What is the perimeter, in inches, of a rectangle with a length of **4** inches and a width of **9** inches?

- A. **13**
- B. **17**
- C. **22**
- D. **26**

ID: 3b44439b



Note: Figure not drawn to scale.

In the figure, line p is parallel to line q , and line t intersects both lines. What is the value of $x + 142$?

- A. 52
- B. 90
- C. 142
- D. 180